

Thanaporn Plangphatthanaphanit (Ruby)

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Work Authorization: F-1 STEM OPT/CPT

PROFESSIONAL SUMMARY

Backend Software Engineer and M.S. Computer Science candidate with 3 years of experience at ExxonMobil building automation tools, cloud monitoring solutions, and distributed systems. Experienced in Python, Java, AWS, Docker, Kafka, and microservices, with hands-on development of cloud-native, AI, and IoT applications.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, PowerShell, Bash

Backend: REST APIs, FastAPI, Node.js, Kafka, Microservices, Distributed Systems, Event-Driven Architecture

Cloud & DevOps: AWS (Lambda, S3, EC2), Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases: PostgreSQL, MySQL, MongoDB, InfluxDB

Frameworks: React, Next.js, Flutter

PROFESSIONAL EXPERIENCE

SWIFT Platform Analyst | ExxonMobil Corporation | Bangkok, Thailand | Aug. 2022 – Feb. 2025

- Developed PowerShell automation tools that reduced DataDog onboarding time by over 80%, streamlining enterprise monitoring deployments and eliminating manual configuration tasks.
- Built internal tooling to automate DNS health monitoring and reporting, eliminating repetitive manual validation processes.
- Supported large-scale cloud monitoring migration from AppDynamics to DataDog by developing scripts, validating integrations, and coordinating deployment activities across infrastructure and security teams.
- Contributed to platform reliability initiatives supporting critical enterprise financial messaging systems.

IT Support Technician - Intern | Ajinomoto Co., Inc. | Bangkok, Thailand | Jun. 2021 – Aug. 2021

- Built RPA workflows that removed most of a recurring daily task and shipped PowerBI dashboards for the support team.

Information Technology Technician - Intern | SS&C Technologies | Bangkok, Thailand | Jun. 2020 – Aug. 2020

- Built a C++ (MFC) internal data-validation tool and owned its full lifecycle: testing, docs, and user onboarding.

TECHNICAL PROJECTS & RESEARCH

Flexi: Educational Robot Research Project | Python, Robotics, Computer Vision, AI/ML | [GitHub: flexi-robot](#)

- Redesigned an educational robot's speech system with dynamic AI-generated voice feedback and built a serverless text-to-speech pipeline using Firebase Cloud Functions and ElevenLabs API, alongside 4+ interactive HTML5 learning activities with adaptive hints for pre-K–Grade 4 students.

Trading Agents Platform | Python, Kafka, Docker, PostgreSQL, Grafana | [GitHub: trading-agents](#)

- Designed a distributed, event-driven trading platform using Kafka, Python, and PostgreSQL, with containerized microservices for autonomous trading, risk management, and market analytics, evaluating latency, throughput, and fault tolerance.

Serverless AI Image Processing Pipeline | Python, AWS, Docker | [GitHub: images-art-transformation](#)

- Architected a serverless AI image-processing pipeline using AWS Lambda, S3, FastAPI, and Docker, integrating DeOldify and Google Gemini for automated image transformation and GitHub Actions for CI/CD.

RxAssist Smart Dispenser | Python, Node-RED, InfluxDB, Raspberry Pi

- Developed an IoT medication dispenser integrating QR/OCR prescription processing, automated scheduling, and motorized dispensing with a real-time MQTT, InfluxDB, and Grafana monitoring platform..

EDUCATION

University of Washington | Tacoma, WA | M.S. Computer Science and Systems | GPA: 3.76 (Expected March 2027)

- **Thesis:** Adaptive ECG arrhythmia detection using explainable AI and multi-criteria model selection
- **Coursework:** Advanced Algorithms, Applied Distributed Computing, Internet of Things, Cryptology

Mahidol University | Nakhon Pathom, Thailand | B.S. Information and Communication Technology | Aug. 2022

- **Senior Project:** Developed VHelp, a cross-platform mobile application (iOS & Android) for individuals with depression, featuring 10+ functionalities. Designed the UX/UI and built the application from scratch.

CERTIFICATIONS

- Azure SC-900, Splunk Certified